

CGCT MODEL 1 REPORT

Collaboration in Content Development

MEQ545 ENGINEERING ECONOMICS

Bachelor of Manufacturing Engineering Technology with Honours
Faculty of Mechanical Engineering · Universiti Teknologi MARA, Cawangan Pulau Pinang
Semester Sesi I 2026/2027 (OCT 2026 – FEB 2027) · Class PEM2433A1

DEVELOPED CONTENT

MEQ545 All-Quiz Tutorial Portal (web application)

10 tutorial sets · 114 questions

Live portal: https://meq545-all-quiz-portal-hub.pages.dev/

PREPARED BY

Muhammad Faris Bin Abd Manap [336994] — Responsible Person (RP) / Content Lead
Muhd Azhar Bin Zainol [336907] — Teaching Team
Industry Panel: En Ahmad Khairul Nizam Bin Bahrinal — Branch Manager Principal, Homefield Real Estate Sdn Bhd

21 October 2026

Centre for Innovative Delivery and Learning Development (CIDL)
Collaborative Group: Collaborative Teaching (CGCT)

COMPLETED CGCT MODEL 1 CHECKLIST

Checklist item	Requirement	Evidence / status
(a) Scheme of work that includes the developed topics	Updated scheme of work showing the topics developed collaboratively	PROVIDED — tuned teaching plan with the Developed Content (CGCT Model 1) column and collaboration block (companion docx/pdf; web app: Section 7.4). Status: complete.
(b) Link / screenshots of the developed content	Access link and screenshots of the developed content	PROVIDED — live portal and repository links (Sections 4.3, 7.4); screenshots: Figures 1–11 (Appendix A). Status: complete.
(c) Evidence of session(s) conducted with students and content	Evidence of sessions conducted involving students and the developed content (e.g., photographs/screenshots)	PROVIDED — student session with content (Figure 3), class roster (Figure 6), submissions with scores and attendance (Figure 7), attendance e-signature (Figure 4) and signed attendance record (Figure 9). Status: complete.

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1. INTRODUCTION

1.1 Background

This report documents the implementation of Collaborative Teaching (CGCT) Model 1 — Collaboration in Content Development — for MEQ545 Engineering Economics, Semester Sesi I 2026/2027 (OCT 2026 – FEB 2027), Faculty of Mechanical Engineering, Universiti Teknologi MARA, Cawangan Pulau Pinang. The Centre for Innovative Delivery and Learning Development (CIDL) Collaborative Group: Collaborative Teaching (CGCT) encourages academics to collaborate — among colleagues and, where relevant, with industry — in the development and delivery of teaching and learning content. Model 1 specifically covers the joint development of content by a collaborative group.

1.2 Objectives of the collaboration

- Develop a complete, syllabus-mapped bank of interactive tutorial sets for MEQ545 (10 sets; 107 questions) covering Chapters 1–7 of the course syllabus;
- Incorporate industry-context scenarios and industry validation through an industry panel (project appraisal and cost decision-making in real estate);
- Deploy the developed content as a digital tutorial and attendance platform (web application) used by students in and outside class;
- Institutionalise reusable, version-controlled content that can be maintained and extended in future semesters.

1.3 Scope and compliance

The collaboration is additive. The official syllabus, Course Outcomes (CO1–CO3), assessment structure and weights are unchanged (Individual Assignment 20%, Group Project 20%, Test 1 30%, Test 2 30%). The developed tutorial sets supplement the scheduled tutorial sessions and are released according to the tuned teaching plan.

1.4 Documentation map against the CGCT Model 1 checklist

Checklist item	Where documented in this submission
(a) Scheme of work that includes the developed topics	Tuned teaching plan — schedule table with the added 'Developed Content (CGCT Model 1)' column (companion document); summarised in Section 5.4.
(b) Link / screenshots of the developed content	Section 4 and the links table (Section 7.4); screenshots in Appendix A (Figures 1–11).
(c) Evidence of session(s) conducted with students and content	Section 6 (usage evidence as at 21 October 2026); Appendix A (Figures 3–7, 9); roster and submissions monitoring screenshots.

The completed Model 1 checklist is reproduced at the front of this report, immediately after the cover page.

2. COURSE AND PROGRAMME INFORMATION

Item	Details
Course	Engineering Economics (MEQ545)
Programme	Bachelor of Manufacturing Engineering Technology with Honours
Faculty / Campus	Faculty of Mechanical Engineering, UiTM Cawangan Pulau Pinang
Semester	Sesi I 2026/2027 (OCT 2026 – FEB 2027)
Class group	PEM2433A1
Credit hours	3 credit hours (14 teaching weeks)
Assessment	100% coursework: Individual Assignment 20% · Group Project 20% · Test 1

	30% · Test 2 30%
Course outcomes	CO1 Explain the basic principles and practice of project management [PO1, LO1] (C2); CO2 Display the principles of the engineering economy to manage process design [PO10, LO10] (A5); CO3 Recognise the need for and independently propose engineering economic related analysis in solving economic problems [PO11, LO9] (C4)
Responsible Person (RP)	Muhammad Faris Bin Abd Manap [336994]
Teaching team	Muhd Azhar Bin Zainol [336907]
Industry panel	En Ahmad Khairul Nizam Bin Bahrinal — Branch Manager Principal, Homefield Real Estate Sdn Bhd (CV: Appendix B)
Live course portals	Tutorial portal: https://meq545-all-quiz-portal-hub.pages.dev/ · Teaching plan web app: https://muhammadfaris-prog.github.io/meq545lessonteachingplan/

3. COLLABORATION STRUCTURE AND ROLES

CGCT Model 1 for MEQ545 is delivered through a three-way collaboration: the Responsible Person (content lead and portal developer), the teaching team (co-development and review) and an industry panel (industry co-development and validation).

Member	Role	Contribution to the developed content
Muhammad Faris Bin Abd Manap [336994]	Responsible Person (RP) — Content Lead	Designed the tutorial framework mapped to the MEQ545 syllabus and course outcomes; developed all question sets, worked solutions and answer explanations; developed and deployed the quiz portal web application; monitors usage; prepares CGCT documentation.
Muhd Azhar Bin Zainol [336907]	Teaching Team — Co-Developer	Co-development and item review; vetting of questions and answer keys; contributes to sets T1–T4 and reviews the industry co-developed sets; facilitates tutorial sessions in which the portal is used.
En Ahmad Khairul Nizam Bin Bahrinal	Industry Panel — Industry Co-Developer / Validator	Co-develops and validates the industry-context scenarios for sets T5–T8 (cost estimation, time value of money, project evaluation, comparison of alternatives), grounded in real-estate project appraisal, feasibility studies, cost optimisation and O&M decision-making.

Coordination: regular coordination meetings and blended communication (in-person, WhatsApp, Google Meet), a shared drive for content sources, and the portal repository as the single source of truth for deployed content.

4. DEVELOPED CONTENT — MEQ545 ALL-QUIZ TUTORIAL PORTAL

4.1 Overview

The developed content is a purpose-built web application — the MEQ545 All-Quiz Tutorial Portal — that delivers all tutorial sets, records student attendance and monitors submissions for the course. The portal is a single-page application (HTML/JavaScript) with a Firebase (Firestore) backend, hosted on Cloudflare Pages and version-controlled in a GitHub repository.

- Student access: student-ID login with a personal 4-digit PIN and self-registration on first use, with per-student progress badges.
- Tutorial delivery: 10 tutorial sets; auto-marked multiple-choice and true/false items; short-answer and long-answer items retained for lecturer review; answer explanations revealed for objective items.
- Attendance: an on-screen e-signature is captured before a tutorial can be started and is stored with the session record.

- Staff dashboard: manage and seed tutorial sets, monitor registered students, monitor submissions with scores, and export attendance records and CSV reports in one click.
- Deployment: hosted web application (no installation), mobile-friendly, with content updates managed through the repository.

4.2 Content matrix (10 tutorial sets / 114 questions)

Set	Tutorial title	Chapter	Question mix	Status (21 Oct 2026)	Attribution
T1	Tutorial 1: Engineering Economy in Everyday Life	Ch. 1	10 MCQ	Active — in use (27 submission records)	Teaching team (RP-led)
T2	Tutorial 2: Chapter 1 – Introduction to Engineering Economy	Ch. 1	8 MCQ · 2 short · 2 long	Active — in use (24 submission records)	Teaching team (RP-led)
T3	Tutorial 3: Chapter 2 – Cost Concepts and Design Economics	Ch. 2	5 MCQ · 3 T/F · 2 short	Active — released Week 4 (14 submission records)	Teaching team (RP-led)
T4	Tutorial 4: Chapter 2 – Cost Concepts (Applied)	Ch. 2	15 MCQ · 2 short	Active — released Week 4	Teaching team (RP-led)
T5	Tutorial 5: Ch. 3 & 4 – Estimation and Time Value of Money	Ch. 3–4	13 MCQ	Developed; scheduled release (W8)	Industry co-developed
T6	Tutorial 6: Ch. 5 – Evaluating a Single Project	Ch. 5	8 MCQ · 2 short	Developed; scheduled release (W11)	Industry co-developed
T7	Tutorial 7: Ch. 5 – Annual Worth Method (5.1–5.5)	Ch. 5	2 MCQ · 3 short	Developed; scheduled release (W11)	Industry co-developed
T8	Tutorial 8: Ch. 6 – Mutually Exclusive Alternatives	Ch. 6	9 MCQ · 3 short	Developed; scheduled release (W12)	Industry co-developed
T9	Tutorial 9: Ch. 6 – Alternatives (Set 2)	Ch. 6	6 MCQ · 6 short	Developed; scheduled release (W13)	Teaching team (RP-led)
T10	Tutorial 10: Ch. 6 – Alternatives (Set 3)	Ch. 6	7 MCQ · 6 short	Developed; scheduled release (W13)	Teaching team (RP-led)
Total	10 tutorial sets	Ch. 1–7	114 questions (83 MCQ · 3 T/F · 26 short · 2 long)		

4.3 Access

Live portal: https://meq545-all-quiz-portal-hub.pages.dev/ | **Repository:** https://github.com/muhammadfaris-prog/meq545-all-quiz-portal-hub

Students access the portal from any device (in class during tutorial sessions and outside class for revision). The staff dashboard is password-protected; assessor access can be arranged on request.

4.4 Quality assurance

All items were internally vetted: the teaching team reviewed questions and answer keys, and worked explanations accompany the objective items. The industry-context scenarios (sets T5–T8) were co-developed and validated with the industry panel. Terminology and notation follow the recommended textbook (Sullivan, Wicks & Koelling, Engineering Economy, 20th Ed.).

5. COLLABORATION WORKFLOW AND ATTRIBUTION

5.1 Workflow

- Step 1 — Topic mapping: the RP mapped each tutorial set to the course syllabus (Chapters 1–7) and the course outcomes (CO1–CO3), using the official scheme of work as the base.
- Step 2 — Content development: the RP drafted the question sets, worked solutions and answer explanations in the portal; the teaching team reviewed the items and contributed content (sets T1–T4).
- Step 3 — Industry co-development: the industry panel co-developed and validated the industry-context scenarios for sets T5–T8 (cost estimation, the time value of money, project evaluation and the comparison of alternatives), drawing on real-estate project appraisal, feasibility studies, cost optimisation and O&M decisions.
- Step 4 — Team extension: the teaching team extended the industry-co-developed framework into additional revision sets T9–T10.
- Step 5 — Deployment and monitoring: sets are released through the portal according to the tuned teaching plan; submissions, scores and attendance e-signatures are monitored through the staff dashboard.

5.2 Attribution

Sets	Attribution	Colour code
T1–T4 (Chapter 1 – Chapter 2)	Developed by the teaching team, RP-led (with team co-development and review)	Blue
T5–T8 (Chapter 3 – Chapter 6)	Co-developed with the Industry Panel (industry scenarios and validation)	Red
T9–T10 (Chapter 6)	Team extension of the industry-co-developed framework	Blue

5.3 Industry panel contribution

En Ahmad Khairul Nizam Bin Bahrinal is Branch Manager Principal at Homefield Real Estate Sdn Bhd and holds a BSc in Mechanical Engineering (Korea University). His expertise covers project management, feasibility studies, cost optimisation, operational and capital expenditure (Opex/Capex) planning, and analytics (Power BI/SPSS), with CAD proficiency. His full CV is at Appendix B. Within Model 1 he contributed and validated industry-context scenarios, including:

- T5 — financing and time-value-of-money settings for property investment appraisal (loan schedules, cash-flow timing, interest computations);
- T6–T7 — single-project evaluation and annual-worth applications using property acquisition, refurbishment and operations cases;
- T8 — comparison of mutually exclusive alternatives with real cost drivers (e.g., refurbishment versus replacement of facilities);
- Validation — review of assumptions, terminology and practicality of the industry questions against current industry practice.

5.4 Tuned teaching plan (checklist item (a))

The official teaching plan for MEQ545 has been tuned to embed the collaboration: the weekly schedule table now includes a 'Developed Content (CGCT Model 1)' column mapping each week to the portal sets (including colour-coded attribution), and a collaboration block (members, industry panel, workflow and attribution legend) has been added. The syllabus, course outcomes, assessment structure and weights are unchanged. The tuned plan (companion document: 'OCT26 MEQ545 TEACHING PLAN (CGCT MODEL 1 – COLLABORATIVE CONTENT

DEVELOPMENT)') is the primary evidence for checklist item (a); a web-app version of the plan is available at the link in Section 7.4.

6. IMPLEMENTATION EVIDENCE (AS AT 21 OCTOBER 2026)

6.1 Usage metrics

Metric	Value
Registered students (Class PEM2433A1)	24
Tutorial sets deployed and active	4 (Tutorials 1–4)
Questions deployed (of 114 developed)	49
Submission records	65 (Tutorial 1: 27 · Tutorial 2: 24 · Tutorial 3: 14)
Unique completions	51 student–tutorial completions across Tutorials 1–3
Attendance e-signature records	90
Average score — Tutorial 1	74.8%
Average score — Tutorial 2	73.4%
Average score — Tutorial 3	83.6%
Highest score	100% on Tutorials 1–3

All metrics are drawn live from the portal database (Firebase Firestore) and are reproducible from the staff dashboard; the screenshots in Appendix A (Figures 5–7, 9) evidence the same.

6.2 Evidence of sessions conducted with students and content (checklist item (c))

Figure 3 shows a live student session: the student is signed in and the portal lists the available tutorial sets with per-student completion badges. Figure 6 shows the roster of all 24 registered students, and Figure 7 shows the staff view of submissions with per-student scores and attendance ticks. Figure 4 shows the attendance e-signature pad presented before a tutorial starts. Figure 8 shows the student question view with a worked explanation, and Figure 9 shows the signed attendance record generated by the portal, in which each student's captured e-signature is printed against the tutorial date.

6.3 Notes on interpretation

Submission record counts (65) exceed unique completions (51) because students may re-sit a tutorial (the portal retains every attempt). Attendance signatures (90) are captured at each tutorial start, including repeat attempts, and serve as the digital equivalent of the tutorial attendance list.

7. REFLECTION AND CONTINUATION PLAN

7.1 Outcomes

Adoption of the developed content has been strong: by Week 4 the portal had recorded 27 submissions (20 unique students) for Tutorial 1, 24 (19 unique students) for Tutorial 2 and 14 (12 unique students) for Tutorial 3, with average scores of 74.8%, 73.4% and 83.6% respectively. Attendance is captured digitally with signatures stored per session, objective items are marked automatically, and students receive instant answer explanations — freeing tutorial time for discussion.

7.2 Observations and improvements

- Short-answer and long-answer items (26 of the 107 questions) are retained for lecturer review and manual marking; a marking view will be added to streamline this.
- Future versions will add per-set time windows and optional randomised question order to strengthen academic integrity.

- Network conditions during class are manageable: the portal is lightweight and works on student data plans.

7.3 Continuation plan

- Release the remaining sets according to the tuned teaching plan: T3–T4 (Week 4), T5 (Week 8), T6–T7 (Week 11), T8 (Week 12), T9–T10 (Week 13).
- Continue industry panel engagement for the applied scenarios (Chapters 3–6) and extend co-development to Chapter 7 (depreciation and income taxes) in a future semester.
- Reuse and maintain the portal for the next cohort; content is version-controlled in the GitHub repository.
- Share the practice as a CGCT example through CIDL channels (CGCT Padlet).

7.4 Links

Resource	Link
Tutorial portal (live web application)	https://meq545-all-quiz-portal-hub.pages.dev/
GitHub repository (source and content sets)	https://github.com/muhammadfaris-prog/meq545-all-quiz-portal-hub
Teaching plan web app (student-facing)	https://muhammadfaris-prog.github.io/meq545lessonteachingplan/
This report — web version	https://muhammadfaris-prog.github.io/meq545-cgct-model1-report/
CIDL CGCT Padlet (collaborative teaching group)	https://padlet.com/cidlcollaborativeteaching/uitm-cg-collaborative-teaching-94dvrcgck0i2droe

APPENDIX A — EVIDENCE: SCREEN CAPTURES

All captures were taken from the live portal (meq545-all-quiz-portal-hub.pages.dev) and the teaching plan web app, and are provided as separate image files in the accompanying evidence folder; Figure 9 is the signed attendance record exported by the portal (full signed PDF supplied in the evidence folder).

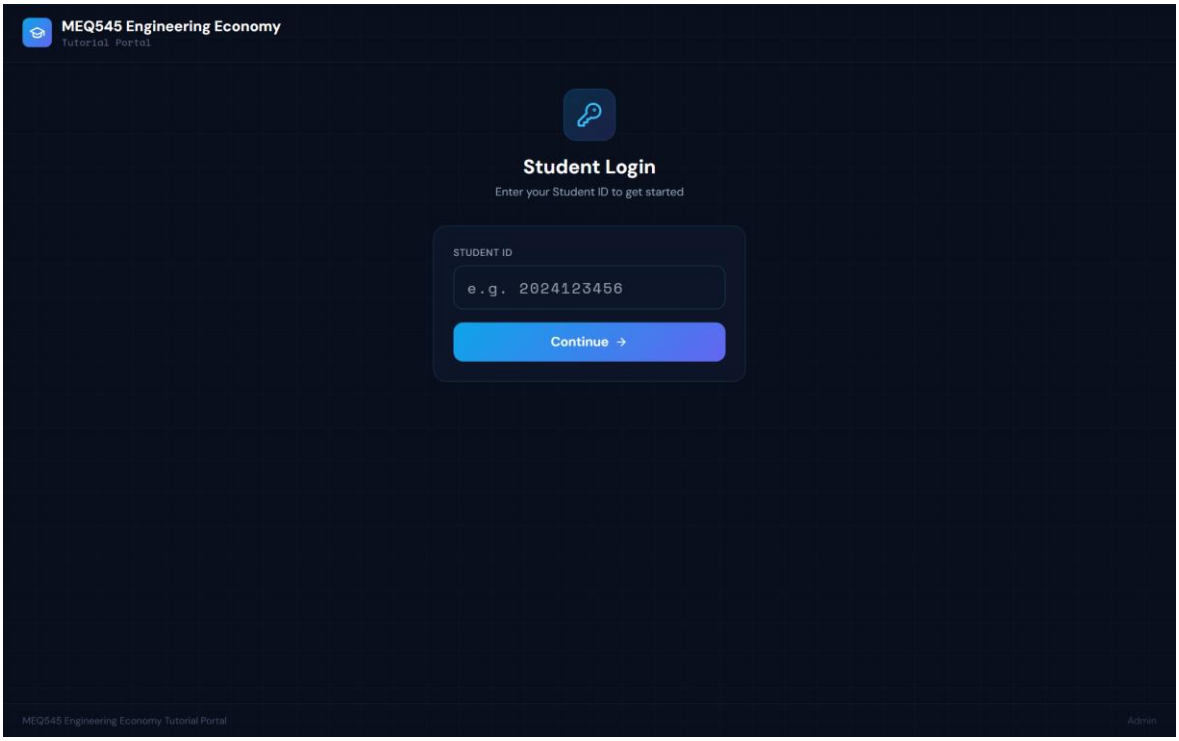


Figure 1 — Student login page of the MEQ545 All-Quiz Tutorial Portal.

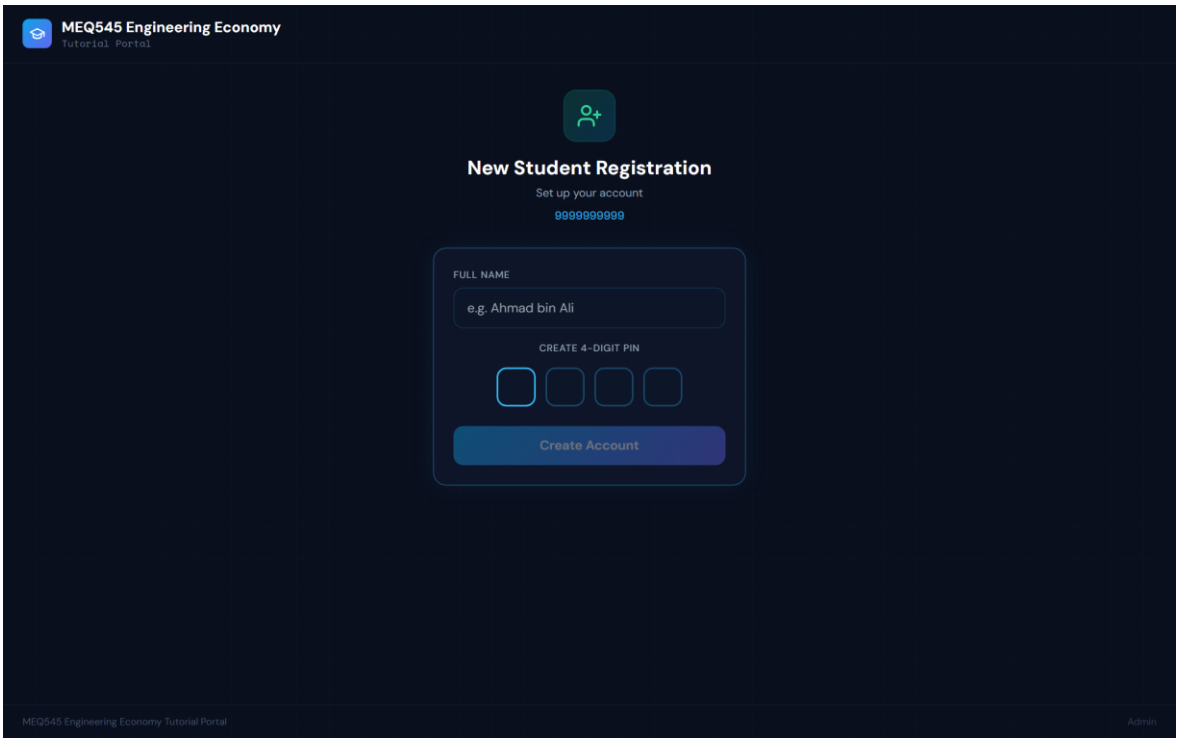


Figure 2 — New student registration (self-service account setup at first login).

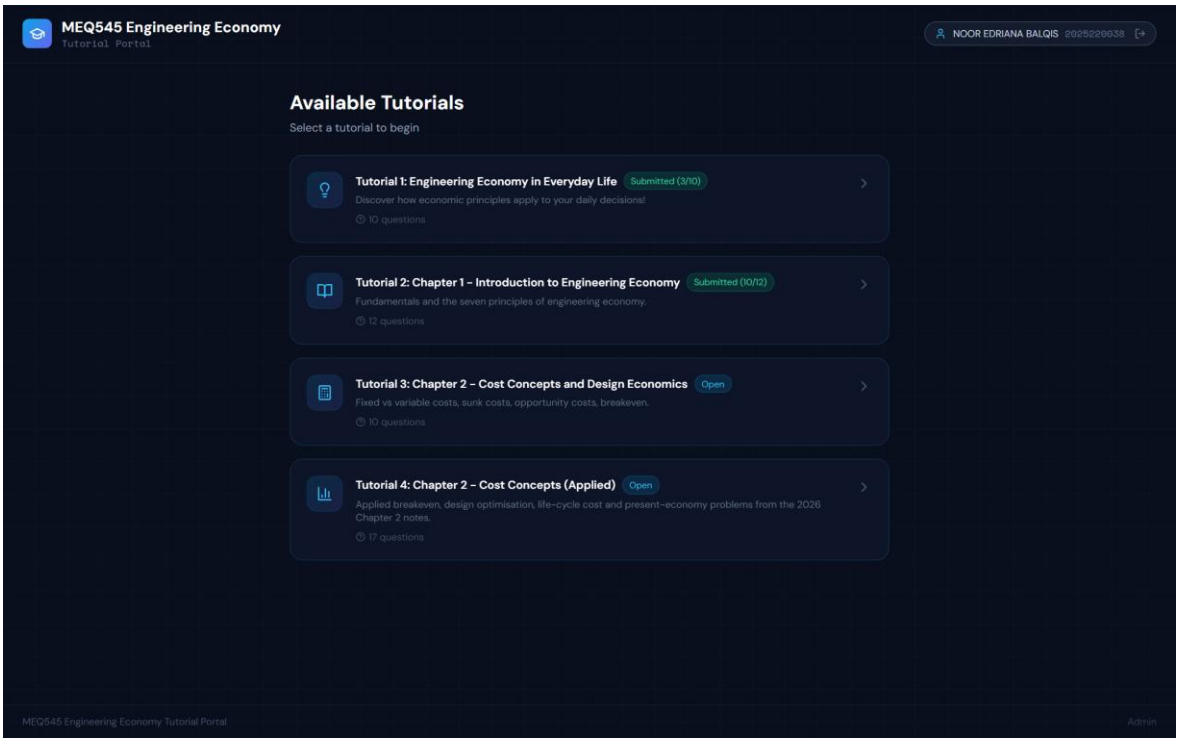


Figure 3 — Live student session: signed-in student badge (name and ID), available tutorial sets and per-student completion badges — session evidence with students and the developed content.

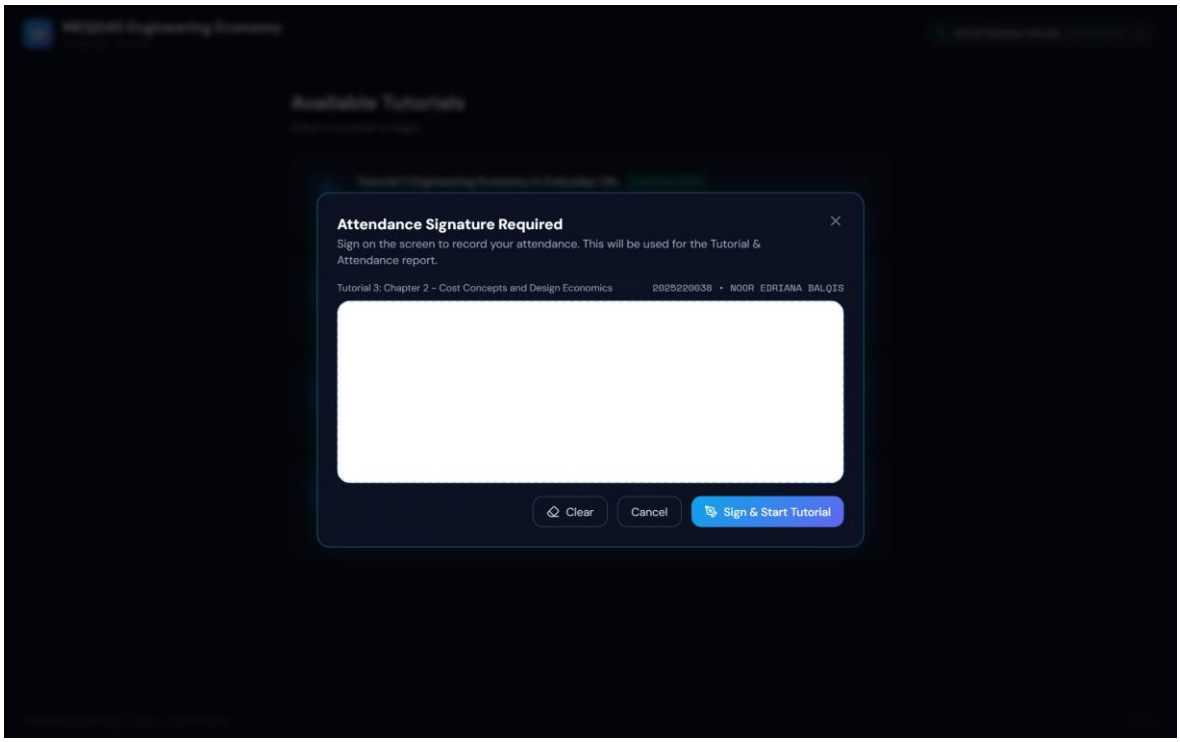


Figure 4 — Attendance e-signature pad presented before a tutorial starts (shown before signing; every captured signature is stored and printed in the signed attendance record — Figure 9).

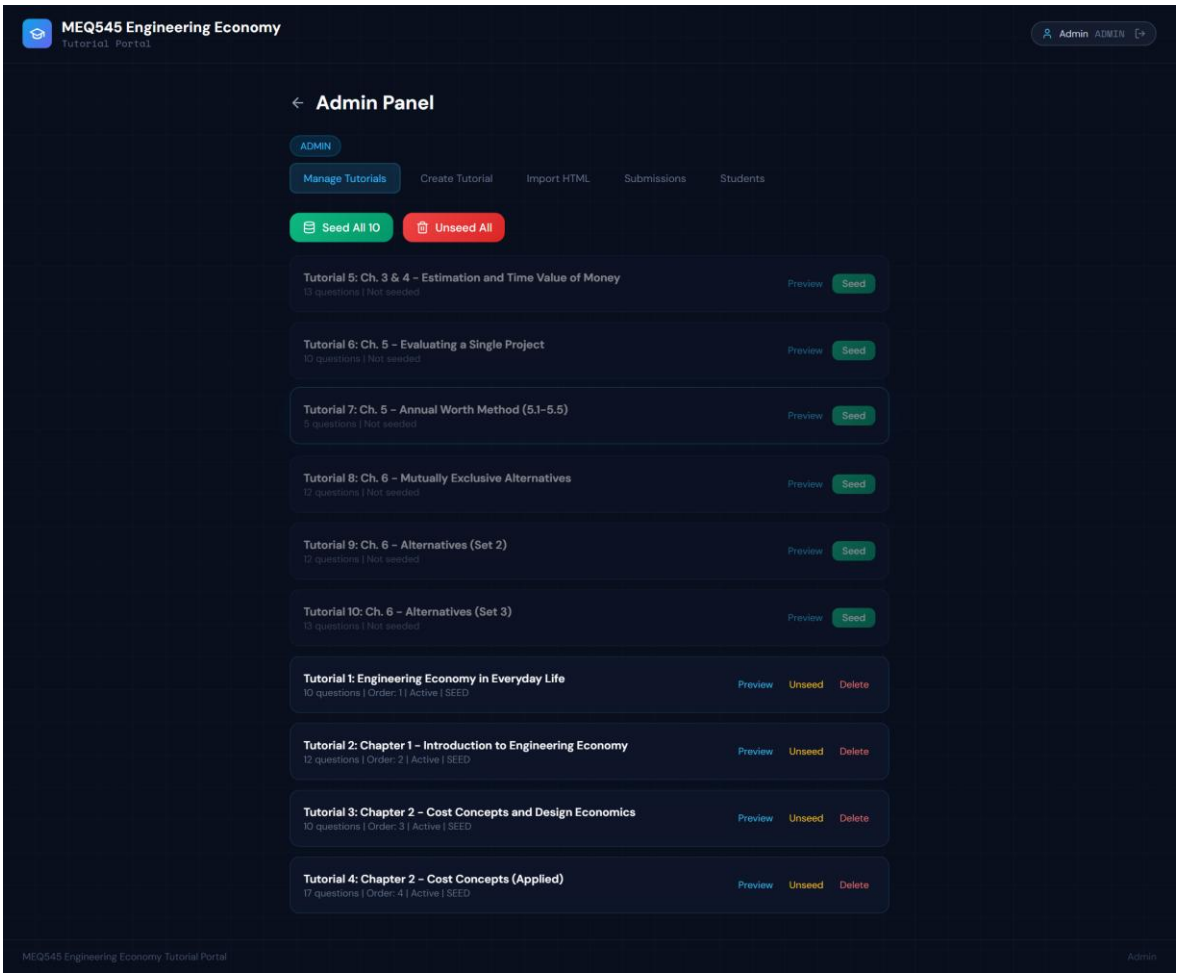


Figure 5 — Staff dashboard: tutorial management (Tutorials 1–2 active; Tutorials 3–10 developed and ready for scheduled release).

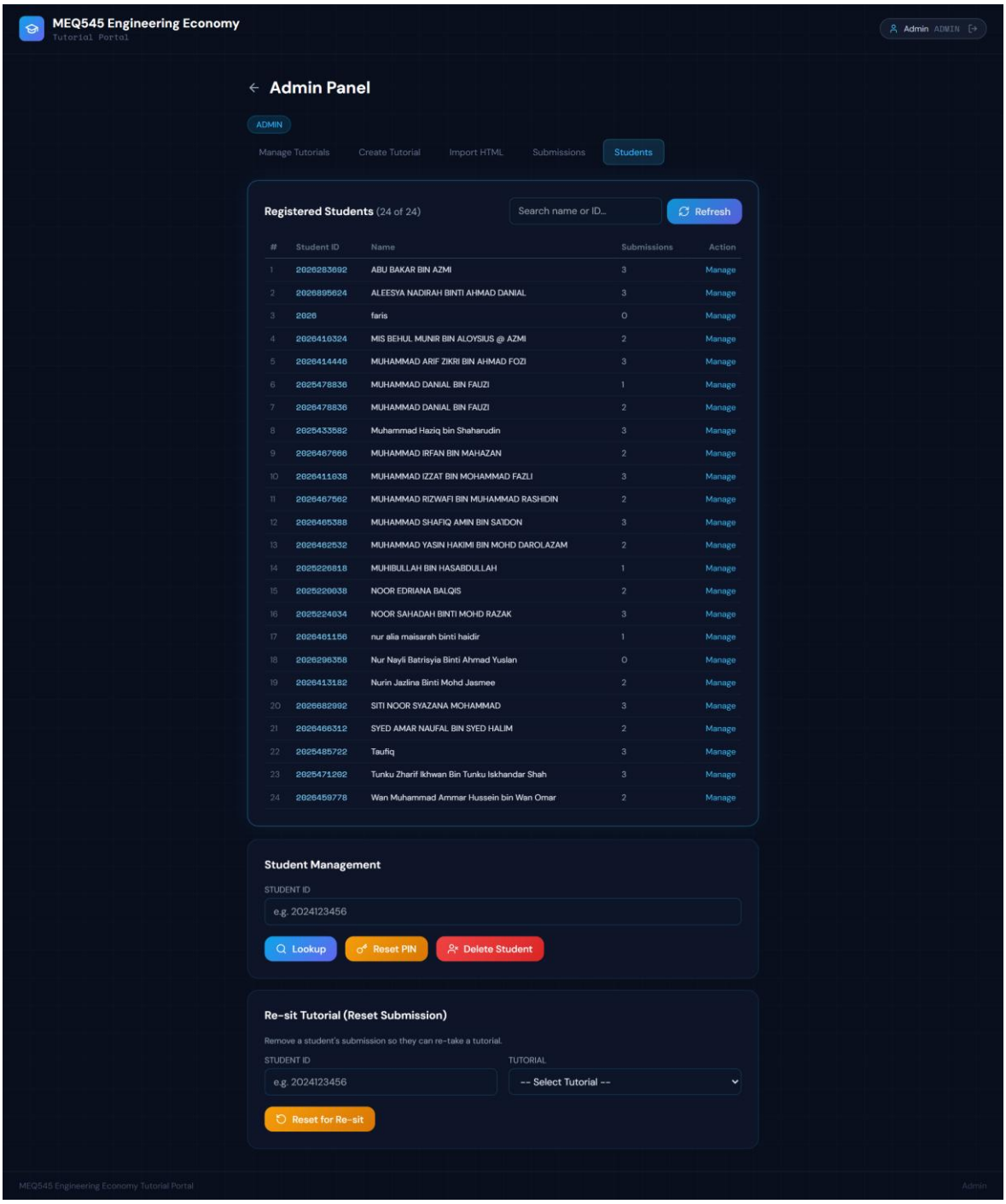


Figure 6 — Registered students of class PEM2433A1 (24 of 24) with submission counts.

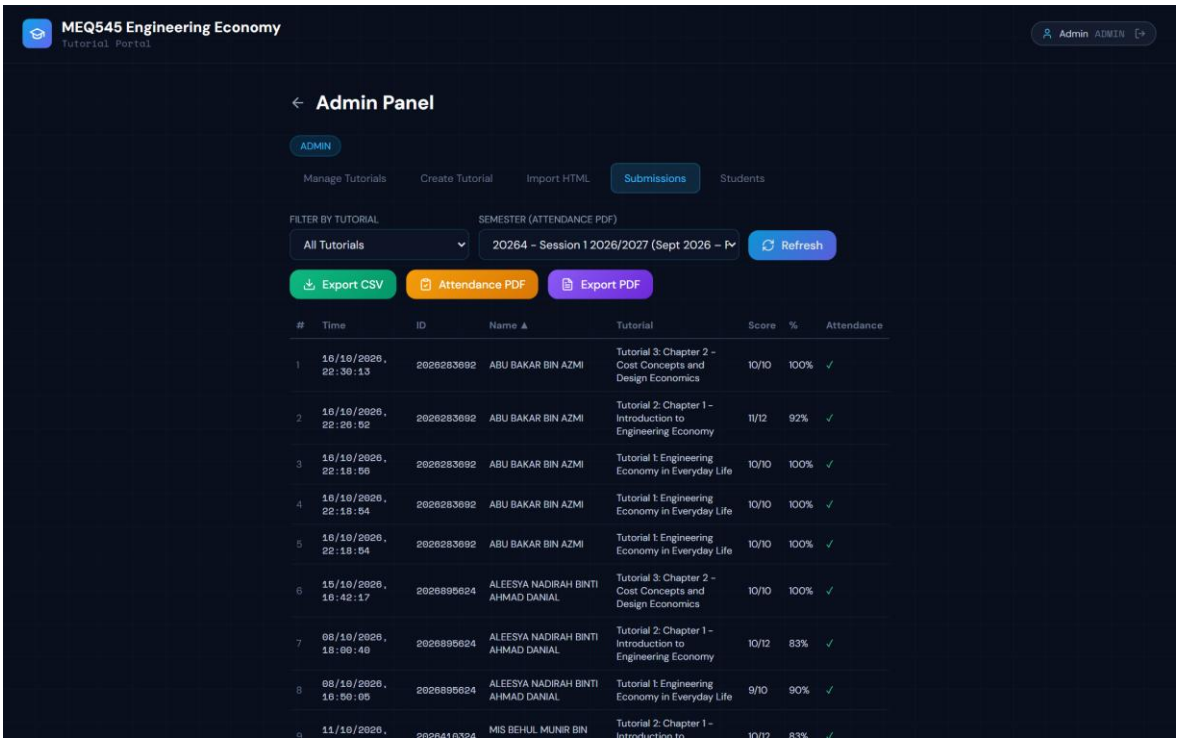


Figure 7 — Submissions monitoring: per-student scores and attendance ticks (extract).

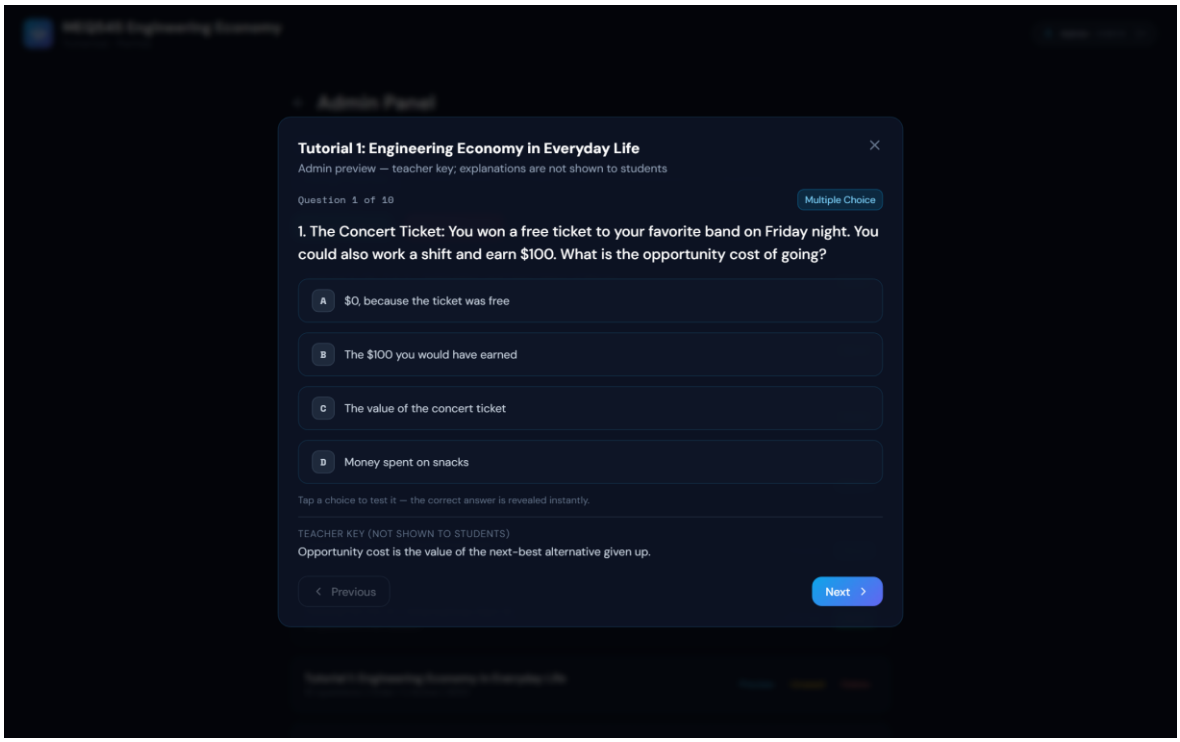


Figure 8 — Student question view (Tutorial 1, Question 1) with answer choices and worked explanation.

LAPORAN : 3.1
Data Sah Sehingga :
21/10/2026

UNIVERSITI TEKNOLOGI MARA

SENARAI KEDATANGAN

20264 - Session 1 2026/2027

HEA/RP/KH-01

NAMA PENSYARAH :
KOD FAKULTI : EM - FACULTY OF MECHANICAL ENGINEERING
KURSUS : MEQ545
KUMPULAN : PEM2433A1
SEMESTER : 20264 - Session 1 2026/2027

BULAN :

Hari	Masa	Bilik

BIL	NO PELAJAR	NAMA	DISABILITY	PRG	BHGN	TUTORIAL & TANDATANGAN		
						TUTORIAL 1	TUTORIAL 2	TUTORIAL 3
1	2026283692	ABU BAKAR BIN AZMI		EM243		16/10/2026	16/10/2026	16/10/2026
2	2026895624	ALEESYA NADIRAH BINTI AHMAD DANIAL		EM243		08/10/2026	08/10/2026	15/10/2026
3	2026	faris		EM243				13/10/2026
4	2026410324	MIS BEHUL MUNIR BIN ALOYSIUS @ AZMI		EM243		11/10/2026	11/10/2026	
5	2026414446	MUHAMMAD ARIF ZIKRI BIN AHMAD FOZI		EM243		11/10/2026	11/10/2026	20/10/2026
6	2025478836	MUHAMMAD DANIAL BIN FAUZI		EM243				19/10/2026
7	2026478836	MUHAMMAD DANIAL BIN FAUZI		EM243		12/10/2026	12/10/2026	
8	2025433582	Muhammad Haziq bin Shaharudin		EM243		12/10/2026	12/10/2026	18/10/2026
9	2026467666	MUHAMMAD IRFAN BIN MAHAZAN		EM243		11/10/2026	11/10/2026	
10	2026411038	MUHAMMAD IZZAT BIN MOHAMMAD FAZLI		EM243		11/10/2026	11/10/2026	20/10/2026
11	2026467562	MUHAMMAD RIZWAFI BIN MUHAMMAD RASHIDIN		EM243		10/10/2026	10/10/2026	

Petikan dari para 2.13.2. Peraturan Akademik Universiti Teknologi MARA 2009. Pelajar yang tidak mencapai kehadiran 80% dari jumlah jam temu untuk setiap kursus tanpa mendapat kebenaran bertulis dari Fakulti/Cawangan/Pusat Pengajian, tidak dibenarkan menduduki peperiksaan bagi kursus. Bagi kursus yang tiada peperiksaan akhir, kerja kursus tidak diberi penilaian.
Pelajar akan diberi Gred F atau gagal dengan status ZZ serta dikenakan denda RM 100.00.

Tandatangan Pensyarah
Nama Pensyarah:

Tandatangan dan Cop Ketua Jabatan

Page 1 of 3

Figure 9 — Signed attendance record exactly as exported by the portal ('Attendance PDF' in the official HEA/RP/KH-01 format): page 1 shows the student roster (names and IDs) and each student's captured e-signature under the tutorial date. The full multi-page PDF is supplied in the evidence folder.

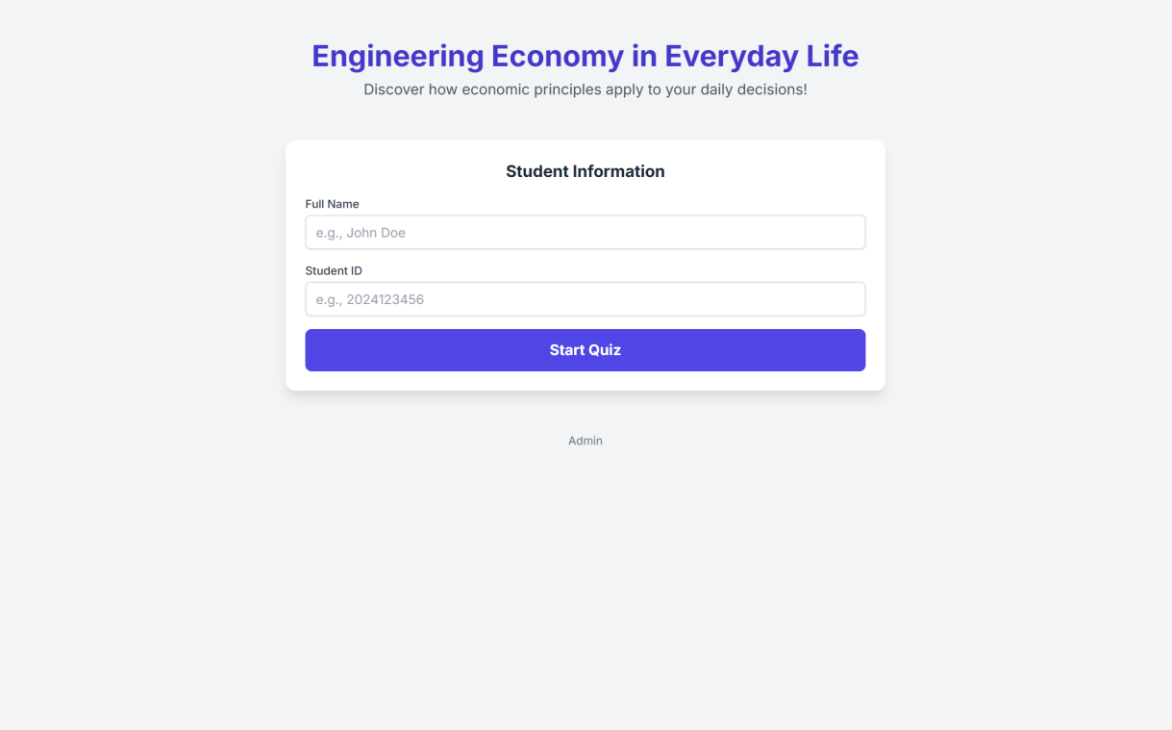


Figure 10 — Standalone tutorial page (Tutorial 1) used for direct-link access.

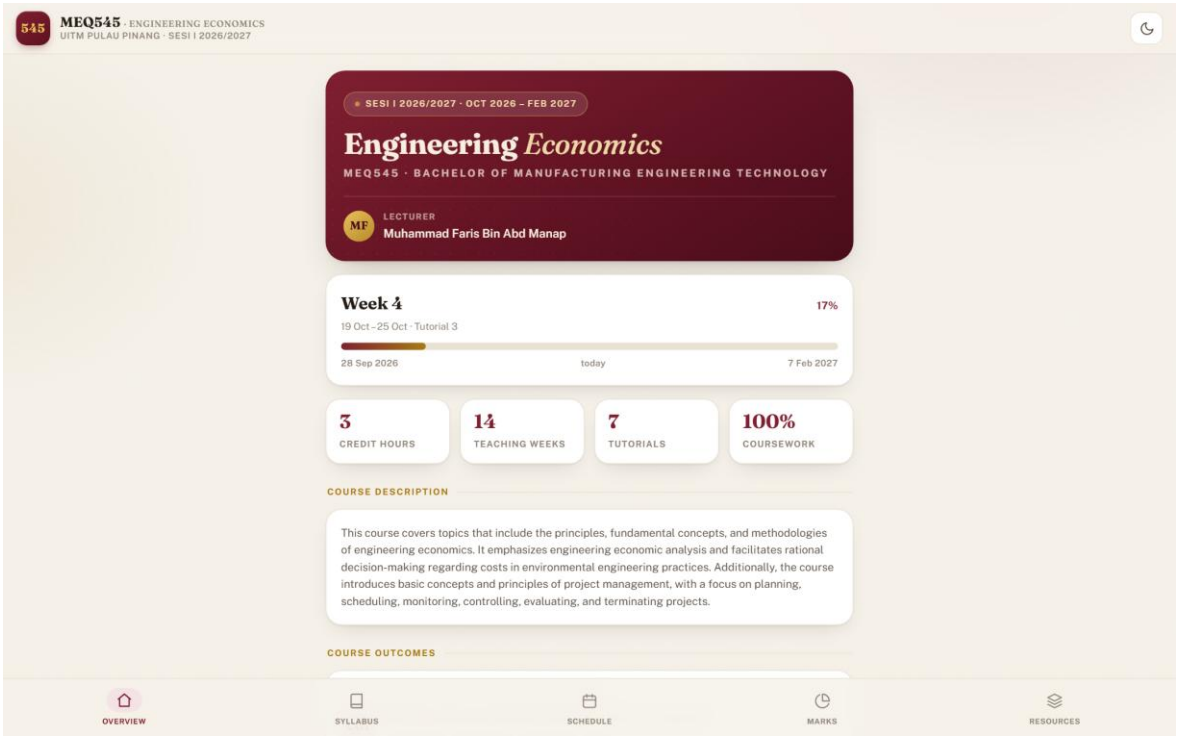


Figure 11 — Teaching plan web app: student-facing view of the tuned teaching plan.

APPENDIX B — INDUSTRY PANEL CURRICULUM VITAE

Curriculum vitae of the industry panel member for this collaboration: En Ahmad Khairul Nizam Bin Bahrinal, Branch Manager Principal, Homefield Real Estate Sdn Bhd.



AHMAD KHAIRUL NIZAM BIN BAHRINAL

Title: Branch Manager Principal, Homefield Real Estate Sdn Bhd
Phone: 013-2969623
Email: nizambahrinal@gmail.com
Website: www.homefield.com.my

SUMMARY

A professional specializes in managing real estate transactions and complex projects management from start to finish. Focus of attention to detail and strategic approach ensure careful budgeting and forecasting. Expertise in OPEX and CAPEX promotes financial stability and organizational growth. By building collaborative relationships with clients and stakeholders, they deliver value-driven solutions that meet both short-term and long-term objectives in a competitive market. Previously working as Site Engineer helping on current situation. This background provided myself with a unique perspective on project development, enabling them to anticipate potential challenges and devise effective strategies to overcome them.

WORK EXPERIENCE

Branch Manager Principal, Homefield Real Estate Sdn Bhd — **June 2023 - Present**

- Spearheaded cross-functional teams to enhance project outcomes and foster collaborative innovation.
- Optimized resource allocation, leading to significant cost savings and improved project timelines. Cost saving 10% last year 2023
- Championed sustainability initiatives, contributing to a more environmentally conscious workplace.
- Developed comprehensive training programs to upskill team members, boosting productivity and morale

Infrastructure & Property Development Strategist, ACEKL, AJC Property Surveyor, AFZ Realty Sdn Bhd — **April 2015 - May 2023**

- Implemented infrastructure designs that improved project efficiency, reducing costs and enhancing regulatory compliance across multiple developments
- Successfully identified and developed high-potential properties, boosting asset value and securing long-term revenue growth for the company

Site Engineer, Lotte E&C Malaysia Sdn Bhd — **June 2013 - Mar 2015**

- Managed full lifecycle of a cutting-edge automation project, meeting all milestones.
- Conducted feasibility studies and risk assessments, mitigating potential project risks.
- Ensure the project run on the site meet all safety and health regulation (ISO, green card, etc)

EDUCATION

Bachelor of Science in Mechanical Engineering — **Mar 2011 - Feb 2013**
Korea University, Seoul, South Korea

- Major in Mechanical Engineering
- Thesis on sustainable energy solutions for urban environments, focusing on innovative technologies to improve energy efficiency and reduce carbon emissions.

Diploma in Mechanical Engineering — **Mar 2009- Feb 2011**
Dongyang Mirae University, Seoul, South Korea

- Relevant coursework in Structural Design and Project Management.

ADDITIONAL INFORMATION

- **Technical Skills:** Automotive Engineering Technology, Project Management, CAD for Structural Engineering, Pro Engineer, SPSS, PowerBI
- **Languages:** English, Malay, Korean.
- **Certifications:** Certificate of Estate Agency, National Valuation Institute, Malaysia
- **Awards/Activities:** Actively participated in project management & giving talk to boost management reliability and company's efficiency

Curriculum vitae of the industry panel member (extract).